



Search in course

Search



English



S



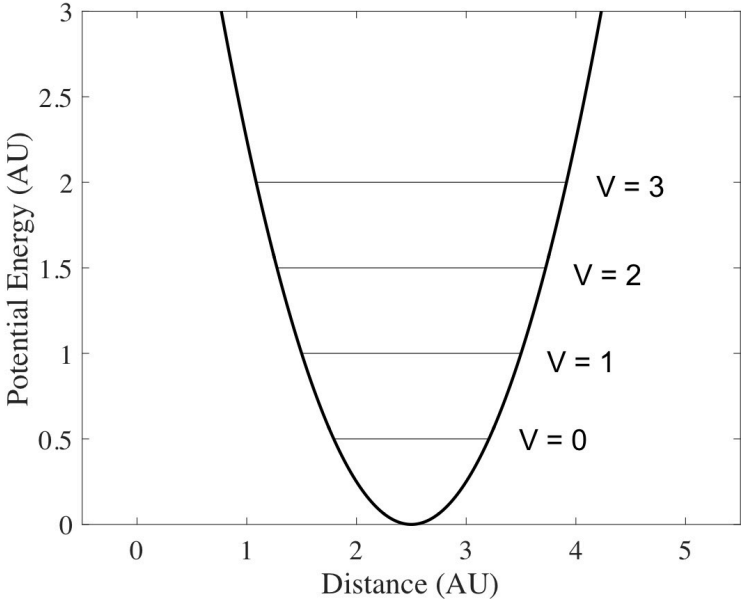
Module 3

Problem 4.14 - January 09, 2023

JD John W. Daily Verified Instructor

2 years ago

Pick a molecule that you can find properties for (i.e. CO) and calculate and plot the ground electronic state vibrational and rotation-level energies. The plot should be like the one below, but also include rotational levels. Include the first five vibrational levels and the first five rotational levels for each ν . Remember you can get the harmonic oscillator constant k from ω_e .



Like Reply 5 Follow this post More ...

5 Replies

Recent

AA Ali Abbas Aslam Learner

2 years ago

(11111,111111111,33333455566)

Like Reply Translate to English More ...

AM Alfredo Maranca Learner

2 years ago

The example shows a parabola, presupposing an harmonic oscillator. As we have seen in the course, the real CO molecule shows a slightly different behaviour, with the right leg of the parabola approaching asymptotically the dissociation energy. For that, there is a curve that gets very near this behaviour, the Morse Potential, that results of Shroedinger equation in another theoretical solution. The Morse potential, however need to be empirically adjusted with real measured values of the constants, as given by <https://webbook.nist.gov/cgi/cbook.cgi?Formula=CO&NoIno=on&Units=SI&Cf=on#ref.25>

The rest potential will be drawn for the $X^1\Sigma^+$ and Morse potential is given by:

$$V(r) = D_e [1 - e^{-(r-r_e)}]^2$$

where r is the potential position and r_e is the internuclear distance (1.20574 Åon NIST site).

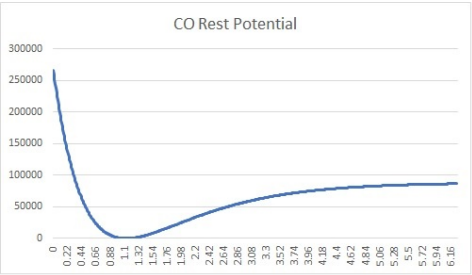
The dissociation energy can be obtained from the values given by NIST site:

$$D_e = \frac{w_e}{4\chi_e} = 6.12 \cdot 10^{-6} cm^{-1}$$

And the morse parameter:

$$a = \sqrt{\frac{8\pi^2 c \mu w_e \chi}{h}}$$

where $\mu = \frac{m_1 m_2}{m_1 + m_2}$ is the reduced mass of the binary molecule. Following the recipe and taking care with these energies in inverse centimeters, because they are tricky, we get:



The ground energy can be obtained by truncating the energy equation for this model in the second term:

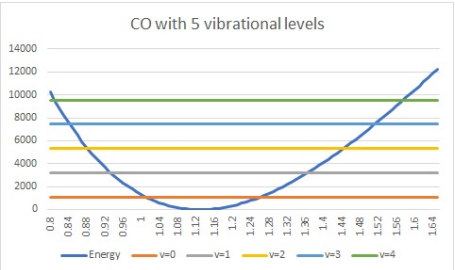
$$G_v = w_e \left(v + \frac{1}{2}\right) + w_e \chi_e \left(v + \frac{1}{2}\right)^2 + \dots$$

$$G_0 = \frac{w_e}{2} + \frac{w_e \chi_e}{4} = 1081.585 cm^{-1}$$

Followed for the next steps:

$$\Delta G_v = G_{v+1} - G_v = w_e - 2w_e \chi_e (v + 1)$$

The result can be seen in a zoomed version of the curve above:



(I'm doing this in excel, sorry for the lines getting out of the parabola. The graphical result can be better in another software)

For the rotational levels, we can sum the rigid rotor term $\tilde{B}$ with the centrifugal stretching $\tilde{D}$

$$\tilde{E}_{total} = w_e + w_e(v + 1) - \chi_e w_e(v + 1)^2 + \tilde{B}J(J + 1) + \tilde{D}J^2(J + 1)^2$$

$$\tilde{B} = \frac{h}{4\pi\mu r_e^2 c} = 1,92 cm^{-1}$$

$$\tilde{D} = \frac{h^3}{4\pi k \mu^2 r_e^4 c}$$

Actually, we don't get direct k ("force constant" or Hooke constant) from the NIST table, but as was said in the enunciate, it can be derived:

$$\nu_0 = \frac{1}{2\pi c} \sqrt{\frac{k}{\mu}} \therefore k = \mu(2\pi c\nu_0)^2$$

We must do a small alteration in D equation to the unities to adapt to the energies in inverse centimeters:

$$\tilde{D} = \frac{h^4}{4\pi k \mu^2 r_e^4 c^2} = 1,94 \cdot 10^{-22}$$

So the entire centrifugal stretching $\tilde{D}$ term is negligible. These levels are lines above each level above drawn but are so close to the main level that will appear invisible in the figure. We have steps of

$$1081 + [3.83, 11.5, 23, 38.3, 57.5]$$

Note that this is not because rotation is not important, on contrary. It's because it's not really quantized in practical treatment.

Like Reply More ...

PR PRITHIKHAR R Learner

2 years ago

harmonic

Like Reply Translate to English More ...

AH Abul Hasan Learner

2 years ago

Determination of force constant of diatomic molecule.
√ If k is the force constant: μ is the reduced mass = $m_1 m_2 / (m_1 + m_2)$: is the fundamental frequency of vibration = frequency of IR radiation absorbed.

2. Identification of functional group.
The presence of functional group can readily detect from their stretching vibrations.
3. Identification of unknown compound. (Qualitative analysis)
This is generally done by comparison of IR spectrum of unknown compound with that of known compound. This comparison is especially made at the finger print region (700 to 1500 cm⁻¹). The spectrum in the finger print region is unique for each molecule.
4. To distinguish between intermolecular and intramolecular hydrogen bonding.
The OH (Hydrogen bonding) stretching frequency does not change on dilution in the case of intramolecular hydrogen bonding. While the intermolecular H-bonding weakens on dilution and there is change in absorption frequency.
5. To study the progress of a chemical reaction (Functional group inter conversion).
As the reaction proceeds the OH stretching band of secondary alcohols slowly disappears and the carbonyl stretching band appear as ketone is formed.

Like Reply More ...

MB M.D ARSHAD BAIG Learner

2 years ago

Like Reply Translate to English More ...